# THE GAS ROUTE
## BURIED PIPES, INACCURATE LOCATIONS AND TEMPORARY DREAMS

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Maps and cartographic diagrams are visual representations used since early ages to identify ownership and power, but as you will see also images able to manipulate collective desires and hopes. In this short text you are about to read, I ask myself, as a simple individual but also as an avid reader of maps and images, how a huge scale distribution network, that of natural gas, could have disappeared in plain sight. To do so, I have scrambled through archives full of images, expired dreams and very low-resolution maps.

In most European countries, natural gas has silently powered homes for nearly fifty years, arriving from an undisclosed location. Today, the political significance of gas infrastructure partly exposed its geographical presence while abstracting it as an intangible entity at various distances and scales. Below ground, a network of pipes crosses borders and territories, molding landscapes and politics, yet vanishing from sight and perception.

The visual materials supporting this text, all found in the archive of Fiera Milano and the one of Europe's largest state-owned oil and gas company, the Italian Eni, give us an insight on the ever-changing narratives around energy: from the celebration of a new resource as an ecological breakthrough to replace other fossil fuels to the present day when the same solution has become already outdated. In a time of facing new energy and environmental crises, this text attempts to bridge with the past and expose a national tendency to forget and hide through the use of images.

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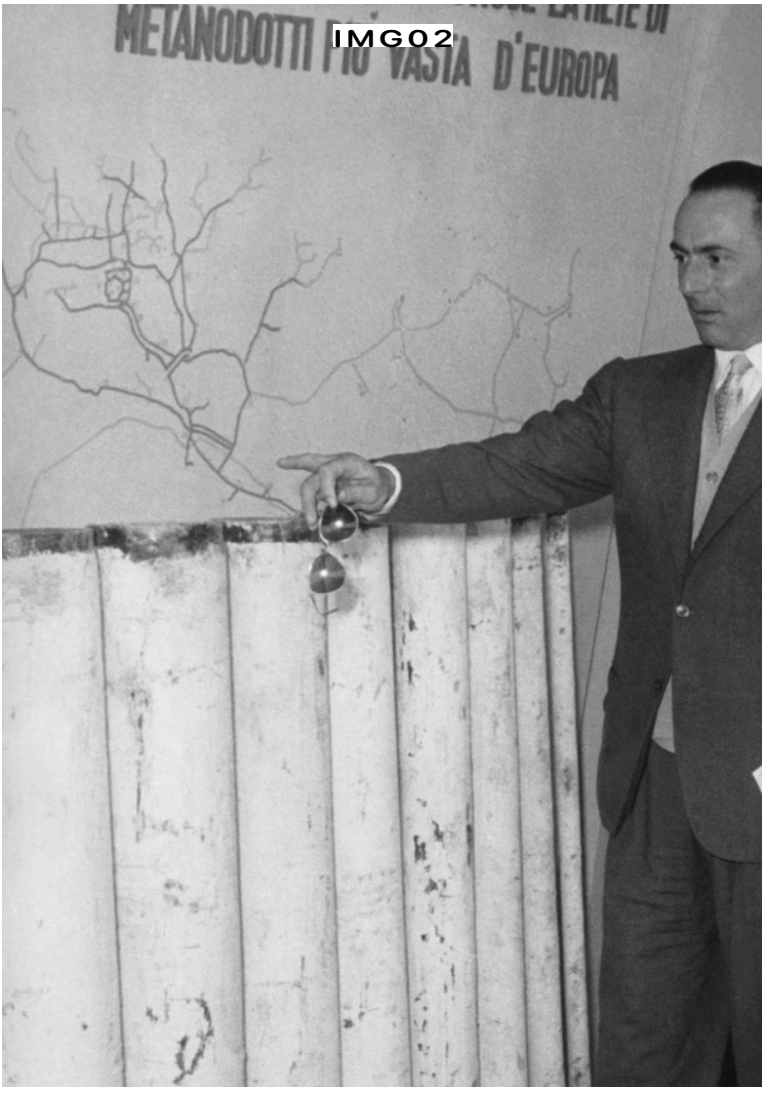
Image Credits 91

# ARCHIVE

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IMG01


METANO
SNAM AGIP
AGIP SNAM
S.N.A.M.
XXVIII FIERA DI MILANO - 12-30 APRILE 1950 - PUBLIFOTO


IMG02
METANODOTTI PIÙ VASTA D'EUROPA

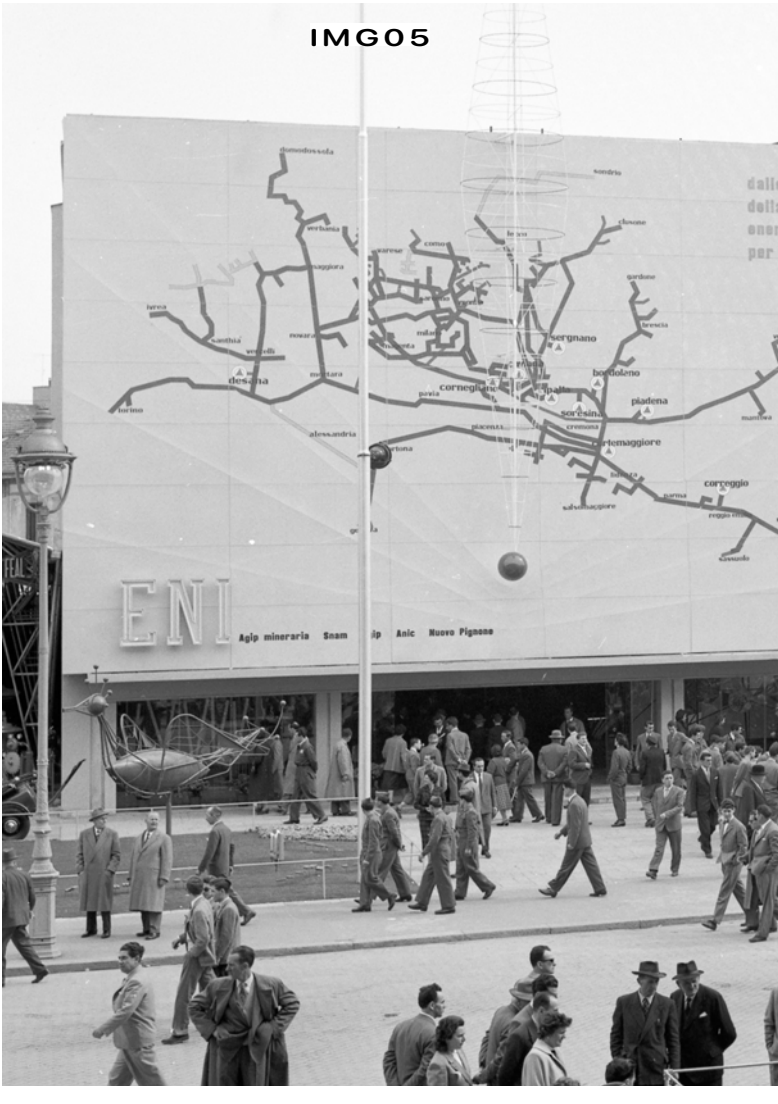
IMG03


DALLE PROFONDITÀ DELLA TERRA
ENERGIA PER IL LAVORO ITALIANO
METANO
SNAM
AGIP

IMG04


E.N.I.
SNAM
dalle profondità della terra
energia per il lavoro italiano
Ingresso

IMG05


ENI
Agip mineraria Snam ip Anic Nuovo Pignone
domodossola
sondrio
verbania
como
lecco
chiavenna
gardone
brescia
sergnano
bergamo
desenzano
cremona
piadena
mantova
alessandria
vigevano
correggio
parma
reggio emilia
salsomaggiore


profondità
terra
gia
il lavoro italiano
metanodotti e derivazioni
totale
giacimenti
allonsine
ravenna
cotignola
santerno


IMG06
E.N.I.
BIELLO

IMG07

SNAM

## Il trasporto degli idrocarburi

La scoperta di importanti giacimenti di metano nella valle del Po ha creato l'improvvisa necessità di provvedere al trasporto di grandi quantità di gas alle industrie di tutta l'Italia settentrionale.

A tale opera si è dedicata la SNAM la quale in cinque anni è riuscita a costruire dal nulla una rete di metanodotti lunga oltre 4600 chilometri, con una capacità di trasporto complessiva di 25 milioni di metri cubi al giorno di gas naturale. Oggi essa ha raggiunto i 5000 km. di sviluppo.

Questa rete è la più lunga e la più moderna dell'Europa occidentale, e una delle maggiori del mondo. La sua costruzione ha richiesto l'impiego di circa 200 mila tonnellate di tubi di acciaio; per la natura dei terreni attraversati (la valle del Po è fittamente abitata) ha imposto che si costruissero 500 attraversamenti di strade di grande comunicazione e 700 attraversamenti di ferrovie, oltre a numerosi grandi attraversamenti aerei di fiumi.

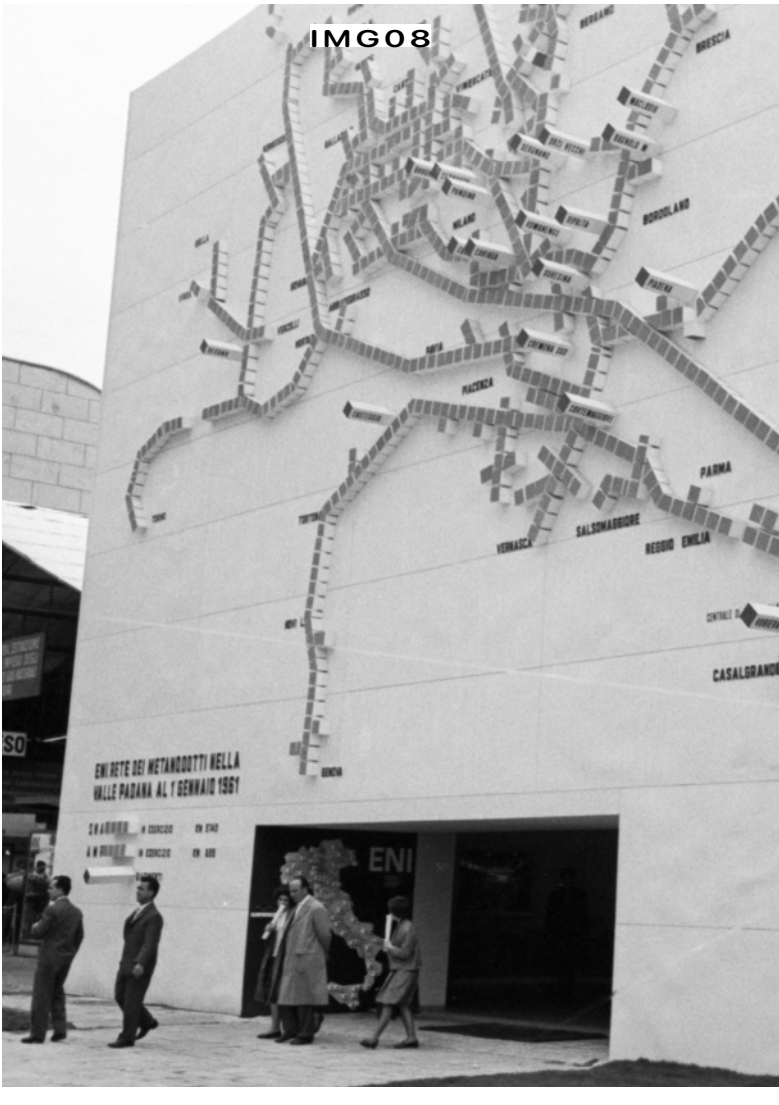
L'esperienza acquisita nel compimento di questa grandiosa opera ha permesso alla SNAM di vincere importanti gare internazionali di appalto, come quella per l'oleodotto Suez-Il Cairo (lunghezza 145 km., diametro 32 cm.).

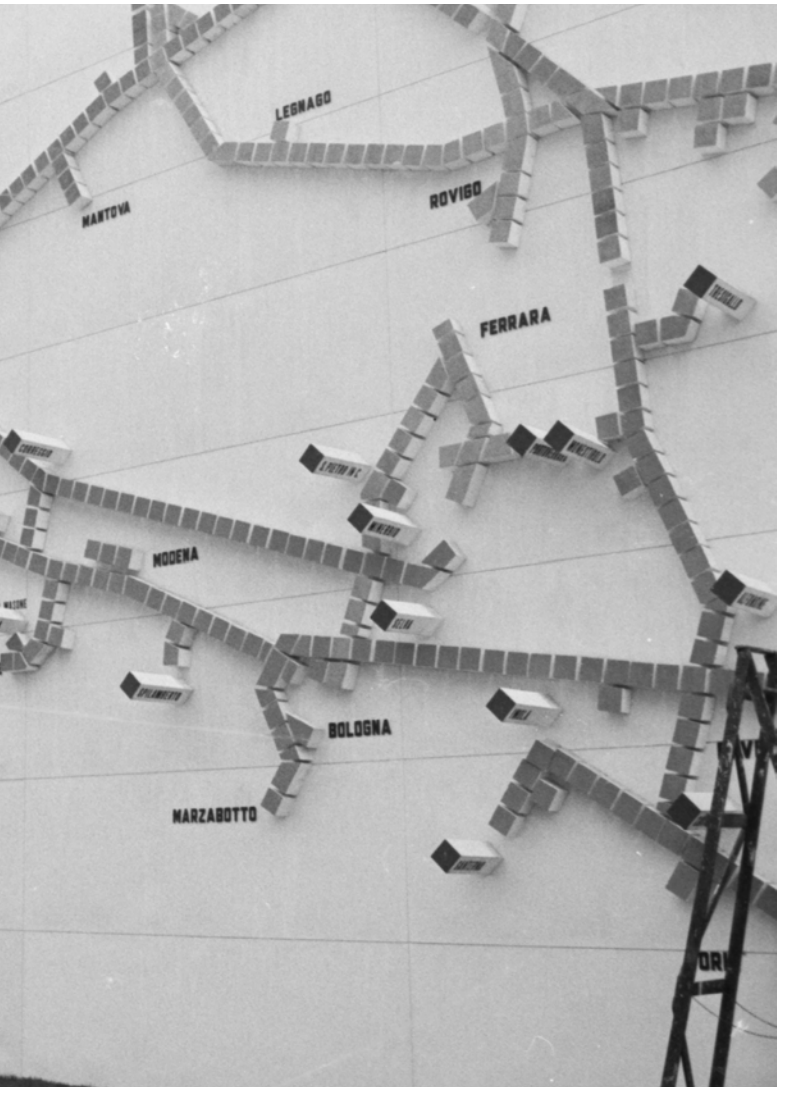
Alle dipendenze della SNAM operano alcune industrie meccaniche, delle quali la più importante è il NUOVO PIGNONE, che costruisce impianti di perforazione, compressori e motocompressori e impianti completi per raffinerie.

◀ L'arabesco degli attraversamenti aerei dei metanodotti sui grandi fiumi padani è ormai parte viva di quel paesaggio

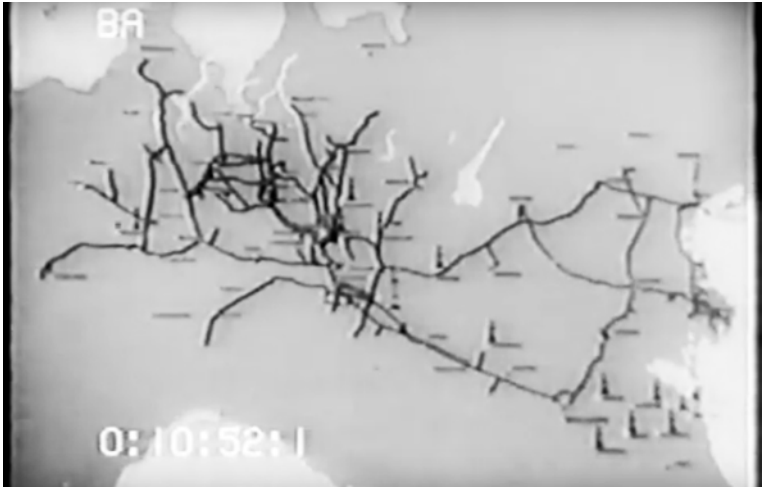
**Rete dei metanodotti «E.N.I.», di circa 5.000 km**




IMG08
BRESCIA
BORGOLAVO
PARMA
REGGIO EMILIA
SALSOMAGGIORE
VERNASCA
CASALGRANDE
ENI
ENI RETE DEI METANODOTTI NELLA
VALLE PADANA AL 1 GENNAIO 1961


LEGNAGO
MANTOVA
ROVIGO
FERRARA
MODENA
BOLOGNA
MARZABOTTO

IMG09




IMG10

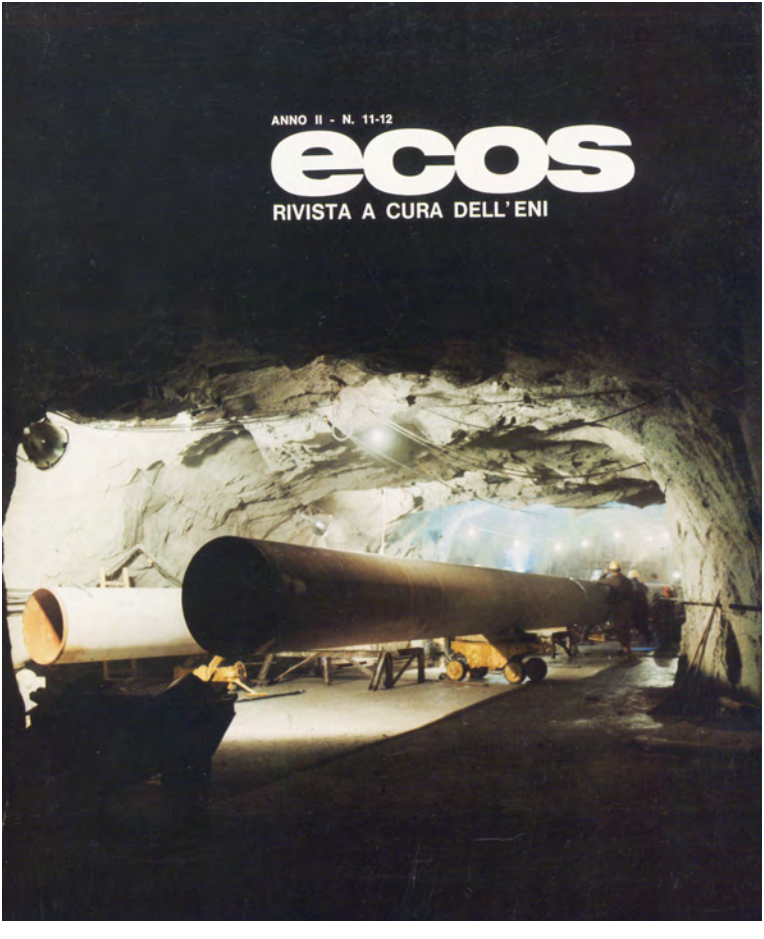
IMG11



# IMG12


Metanodotti in esercizio
Metanodotti in costruzione
Metanodotti in progetto

IMG13



IMG14



# IMG15



IMG16



IMG17




IMG18

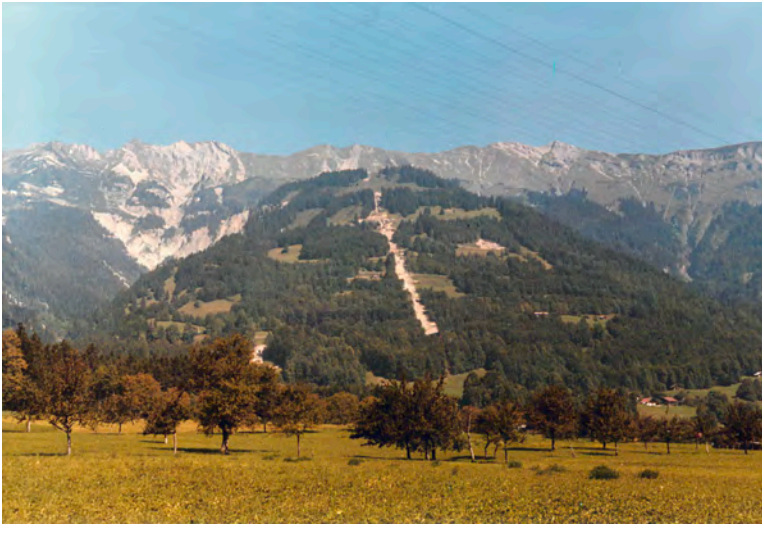
IMG19



IMG20



IMG21




IMG22

IMG23



## 2. *La SNAM entra nella scena internazionale*

Altri Paesi europei già importavano metano. La Germania aveva iniziato le importazioni dall'Olanda, con quantitativi significativi per il mercato interno, nel 1967; la Francia, con modesti quantitativi di gnl algerino, dal 1965 e poi, dal 1967, col gas olandese; il Belgio, vicino al confine olandese, aveva pure iniziato le importazioni da questo paese nel 1967; l'Inghilterra importava metano algerino dal 1965.

Le aree di consumo di questi paesi erano molto prossime alle località in cui erano ubicate le riserve allora disponibili (Olanda e URSS) per la commercializzazione internazionale.

Per l'Italia, chiusa al Nord alle Alpi, esistevano notevoli difficoltà da superare sia sul piano strettamente tecnico sia su quello economico-finanziario. Rispetto agli altri paesi si sarebbero dovuti sostenere più elevati costi di trasporto per portare metano dal Nord Europa e dall'Est europeo fino alle nostre aree di consumo.

Anche per l'accesso alla fonte di approvvigionamento di gas naturale liquefatto dalla Libia, nonostante la relativa vicinanza al nostro Paese, si trattava di uscire da livelli di approvvigionamento sperimentali, quali erano quelli già operativi tra l'Algeria da una parte, Francia e Inghilterra dall'altra. Infatti questi due paesi importavano rispettivamente 0,5 e 1 miliardo di metri cubi all'anno con impianti di modeste dimensioni realizzati con tecniche non ancora sofisticate.

L'importanza e la significatività dell'accordo, firmato nel

Tab. 1

EUROPA: CONSUMO INTERNO LORDO DI GAS NATURALE
*(milioni di metri cubi da 9.100 Kcal)*

| Anno | Italia | Germania | Francia | Belgio | Olanda |
|---|---|---|---|---|---|
| 1960 | 6.447 | 801 | 2.936 | 65 | 346 |
| 1961 | 6.832 | 871 | 4.131 | 65 | 469 |
| 1962 | 7.150 | 1.127 | 4.876 | 67 | 515 |
| 1963 | 7.267 | 1.420 | 5.012 | 65 | 596 |
| 1964 | 7.635 | 2.068 | 5.244 | 65 | 822 |
| 1965 | 7.741 | 2.968 | 5.035 | 77 | 1.589 |
| 1966 | 8.452 | 3.609 | 5.270 | 139 | 3.052 |
| 1967 | 9.232 | 4.688 | 6.003 | 531 | 5.512 |
| 1968 | 10.791 | 7.748 | 7.225 | 1.291 | 9.091 |
| 1969 | 11.925 | 10.923 | 8.802 | 2.656 | 13.362 |
| 1970 | 12.954 | 15.471 | 10.043 | 4.173 | 18.838 |
| 1971 | 13.250 | 20.493 | 11.900 | 5.755 | 24.371 |
| 1972 | 15.373 | 26.451 | 14.091 | 7.256 | 31.534 |
| 1973 | 17.373 | 32.921 | 16.522 | 8.778 | 34.805 |
| 1974 | 19.515 | 39.538 | 17.132 | 10.221 | 37.493 |
| 1975 | 22.000 | 41.697 | 18.547 | 9.955 | 38.862 |

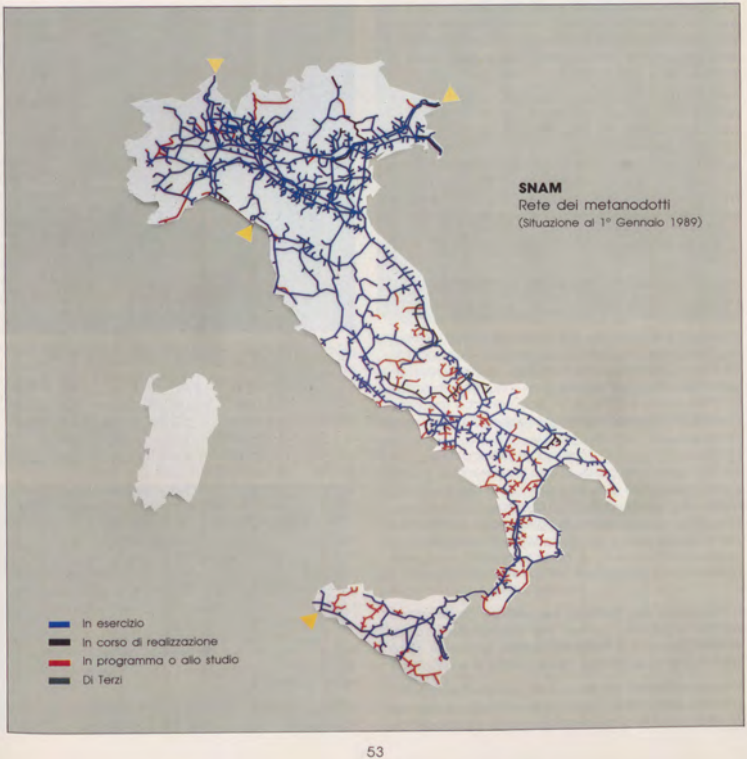


IMG25



IMG26




SNAM
Rete dei metanodotti
(Situazione al 1° Gennaio 1989)
In esercizio
In corso di realizzazione
In programma o allo studio
Di Terzi

IMG27

**Fonte: Snam**



IMG28

Fonte: Snam



Fonte: Snam



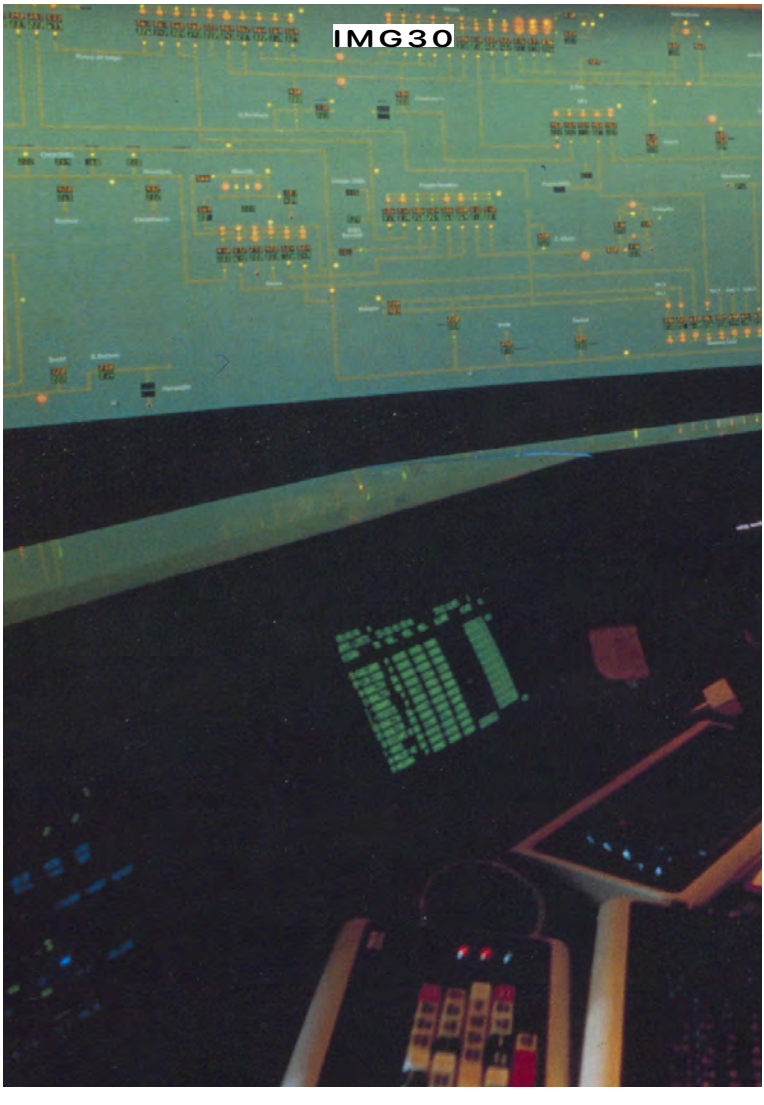

**Il metano è una risorsa esauribile. Per fortuna le riserve europee di questa fonte sempre più preziosa sono notevoli. Può indicarci qualche cifra?**

In Europa è localizzato il 45% delle riserve mondiali di gas naturale. Includendo nell'area europea anche l'Algeria, che è collegata con il sistema europeo del gas attraverso il nostro gasdotto «Transmed», e, nei prossimi anni anche con il costruendo gasdotto che porterà il gas in Spagna, le riserve ammontano a oltre 62 mila miliardi di metri cubi, dei quali 53 mila localizzati in Unione Sovietica.

Come si vede, di metano ne abbiamo in abbondanza. La sfida riguarda il trasporto dai giacimenti ai luoghi di consumo. Ormai le infrastrutture esistenti sono saturate. Per i nuovi, crescenti fabbisogni occorre potenziarle e crearne delle nuove. Alcune iniziative sono già state avviate, il potenziamento del «Transmed», in cui l'Eni è coinvolto direttamente, la realizzazione di nuove condotte sottomarine che collegano i giacimenti del Mare del Nord con il Nord Europa, la realizzazione del gasdotto maghrebino occidentale che, attraverso il Marocco, collegherà i giacimenti algerini alla Spagna. Si tratta di impegni che richiedono un notevole sforzo finanziario, economico, tecnologico e anche politico.

**La sua nomina a presidente dell'Igu è indubbiamente un riconoscimento alla sua persona, al suo lungo impegno per lo sviluppo del metano. Ma rappresenta anche un riconoscimento al ruolo svolto in questo settore dall'Italia e dalla Snam...**

È fondamentalmente un riconoscimento al ruolo che ha avuto il nostro Paese nello sviluppo del gas, un ruolo che Italia e Snam hanno portato avanti in condizioni non facili con notevole successo.


IMG30

IMG31




IMG32

IMG33



IMG34



# IMG35



# IMG36




European Hydrogen Backbone Initiative 2022, supported by Guidehouse

# INTRODUCTION

In the past two years, natural gas and its infrastructure have been mentioned and discussed more than usual in a wide range of media channels. A "usual" scenario is when this infrastructure remains unseen, buried one and a half meters underground beneath cultivated fields, roads, and rivers spanning the globe.

In this short text you are about to read, I ask myself, both as a simple individual and as an avid reader of maps and images, how a huge scale distribution network could have disappeared in plain sight. To do so, I have scrambled through archives full of images, expired dreams and very low-resolution maps.

In January 2021, the Russian invasion of Ukrainian territories sparked immediate attention to this topic in newspapers, news broadcasts, and similar media outlets. This event garnered significant focus primarily due to its potential threat to the entire European natural gas supply system. Subsequently, in September 2022, the Nord Stream 2 pipeline connecting Russia and Germany beneath the Baltic Sea was blown up, filling global media with a single photo, and a thousand other versions of it, from slightly different angles and zooms. The subject of this photo is the bubbling of a stretch of sea, without showing any pipes or anything else we associated before to "infrastructures". The photo does not directly depict what it talks about. Instead it frames its consequence, making it easily incomprehensible and decontextualisable. As if a

submarine monster had exhaled its last breath, the image used to represent this event perfectly translates the overall understanding we have of infrastructures and their impact: distant, not geographically located and abstract.

We, citizens-consumers, threatened on our daily habits by these events, realized all of a sudden how gas is a key component of our lives, embedded in homes and their basic commodities; We realized that we are nodes in a huge network (one of the many), to which we are almost all connected and of which we cannot do without. But even though gas is an issue in the spotlight today, it is not the infrastructure and its political agenda, its history and its geography that are exposed, but only the direct effects it has on our lives, which we may soon forget. The gas distribution infrastructure we have in mind is a service built into houses that disappear when we look for it outside of the window. It's perception is almost like a magical apparition that we now demand and take for granted.

**But how this huge gas infrastructure has been made invisible over time? How has this historically become intelligible to us, the "citizen-consumers", if it has always been invisible?**[1] Through these long pipes whose existence we have recently rediscovered, natural gas is transported from places of extraction to places of consumption, homes or industrial centers, passing through intermediate nodes.

[1] Parks L. (2013). Earth Observation and Signal Territories: Studying U.S. Broadcast Infrastructure through Historical Network Maps, Google Earth, and Fieldwork.

The inherent invisibility of this system of tubes gives news media a representational issue to solve. By doing a simple online research regarding gas infrastructure, what pops up are a few images displaying simplified geographical diagrams of the pipelines routes **IMG34** ■ alongside dozens of stock-like-photos of domestic use of gas such as cooking stoves or heating systems. On the one hand the diagrammatic images of these networks are the main way we understand this object and how we make sense of it. These, however, are very general, and the scale, detail and elements of these maps remain reductive. On the other hand, the huge scale of gas pipelines makes them hard to comprehend even when experienced in physical proximity. You can see them in the landscape, for instance as signposts that mark territory **IMG31** ■ **IMG32** ■. Still, even if providing a high geographical accuracy, these material parts of the infrastructure lack the ability to position the single place within the bigger picture. These make the geography of the network abstract and with it its perception.

Since my curiosity lies in understanding the immateriality of such a tangible and big scale infrastructure, and the inability to have experience of it, I first cross-referenced what I saw on these partial cartographies with what I could see from Google Maps and finally reached some of those places, almost as an act of measuring, witnessing and experiencing the infrastructure with my body. I followed the marks left on the surface to signal the pipeline position, yellow posts fixed in the ground. I traveled along with the gas for a few hours, searching for one yellow stake after another, coming to the conclusion that even though I finally knew where a particular pipeline was, its persistent invisi-

bility perhaps also laid in why I was looking for it, in the history and processes by which it had ended up under the groun there instead of elsewhere; layers of meaning that the yellow poles by themselves could not tell, but maybe the network's maps could hide behind their design and use.

This text explores how the natural gas infrastructure has been perceived in Europe through time and consequently identifies some key moments when it started disappearing from perception. By using visual representations of the gas distribution network from a specific country, Italy, one of the first to develop it in Europe, the different chapters reflect on the political evolution around the issue of gas in many other European countries: from the celebration of this new resource as an ecological breakthrough to substitute other fossil fuels to today's times when gas eventually became a key resource we depend on and that we already find outdated. As it happens, I wrote this text as an expat based in The Netherlands but born and raised in the foggy lowlands of northern Italy, both key places to tell the beginning of the story of natural gas in Europe, and then increasingly secondary locations, almost relics of a blurry forgotten history.

**Maps and cartographic diagrams, representations used since early ages to navigate and identify ownership**[2], allow us to compare what we see with levels of meaning invisible to the eye: political, thematical, morphological, social. Yet cartography has often belonged

[2] Drucker J. (2014). Graphesis: Visual Forms of Knowledge Production. MA: Harvard UP.

to those in power, and as we shall see, it has also been used for promotional and communication purposes, manipulating the view. This text, written from the partial yet specific point of view of a graphic designer, seeks to understand why the gas infrastructure usually comes with an uncertain and imprecise geographical perception by delving into its maps and diagrams drawn at specific historical moments. From these found materials, the geographical space of the infrastructure of gas acquires a new significance; some locations emerge as key places of the early industrial usage of this natural resource; some trajectories inform us about import strategies and, therefore, lack of internal, domestic production; others about the overlap of gas together with other energy resources such as oil. By augmenting the spaces of the current gas routes with historical images and texts, this book has served as the base research behind a travel along a pipeline connecting Italy and The Netherlands. The slow travel, an embodied exploration, was made after writing this book to develop new imaginaries around this subject as an object of power and concern that usually stays invisible. With this "empirical" approach in mind, the blurry history of pipes that is buried with them under the ground has been re-ordered into these pages from a carto-graphic perspective.

# CARTOGRAPHY AS TIME RECORDING DOCUMENT

The gas network is a very extensive tangle made of underground tubes that stretch from one corner of a continent to the other, connecting, in the case of the European one, Siberia to Morocco, Algeria to the North Sea, Turkey to England. These routes cross different nations and their borders, industrial cities and rural places, mountains and glaciers, private property and public places, connecting distant locations that we would only reach by plane. It is an alternative network to the many developed by humans-for humans to navigate such as roads, railways, airways, hiking trails and cycling routes. But this one is not meant to be navigated by people but exclusively by a natural resource in its liquid and gaseous state, brought directly from underground to where it is needed. But where is this resource needed? **The direction of a pipeline itself unveil the hidden connection between places and interests.**[3]

A mountain trail, though seen today as a place of entertainment, was originally created for economic reasons of exchange of goods and movement of people. Its direction and trajectory are reminiscent of these interests when it crosses passes rather than reaching a high peak. Similarly, the places where a pipeline takes space can tell

[3] Dodge, M., C. Perkins. (2010). Maps, Memories, and Manchester: The Cartographic Imagination of the Hidden Network of the Hydraulic City.

us the meaning of some trajectories built 60 years ago.

Throughout history, this infrastructure has been represented with cartographic networks, the best way we found to understand and have an overview of this wide system of tubes. Contrary to what we may think, the representations of this infrastructure were more common to find in the past, albeit in the forms of advertisements or promotional images. **Today, the geography of gas transportation and, in general, of critical infrastructure networks become decreasingly accessible by the publics**[4]. In the rare case when it is accessible, usually it is in the format of official and bureaucratic documents to which we are usually indifferent. While most consumers are familiar with the final outputs of gas within their homes, such as cooking stoves or heating systems, few are used to knowing the network infrastructure and its localized sites that make this possible. Can we shift critical attention from everyday objects of consumption to the infrastructure that supports that system and draws new geographies of power and dependence?

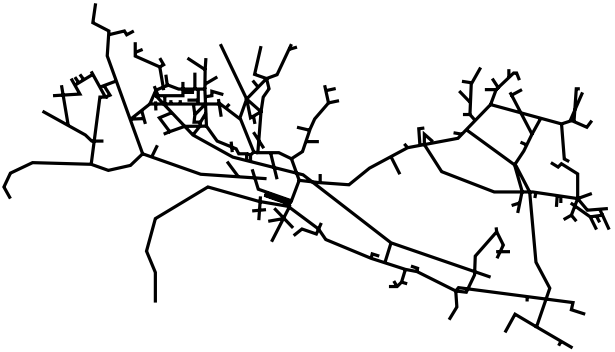
**When trying to understand certain subjects, usually of unreachable physical or conceptual dimensions, direct experience through our bodies is not enough**[5], **so we rely on scaled representations such as diagrams, maps, charts, sequences of fragments, which can translate that incomprehensible object into human time and space, reducing it to a drawing, a series of lines, dots**[6]. We have been doing this as a human

[4] Dodge, M., C. Perkins. (2010).
[5] Drucker J. (2014).
[6] Dodge, M., C. Perkins. (2010).

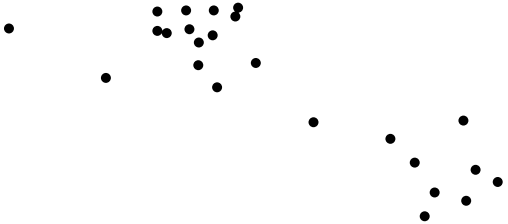
species since ancient times, but today, immersed in a world of "data" that rises to objectivity, they become key to comprehending the world. These visual representations, reducing the bodily dimension of understanding to an abstract mental gaze, give us a new sense of control over what we are talking about. In translating from real to representation, however, several choices are made about what to keep and what to remove, in order to reduce complexity, which can totally change the reading or convey specific messages.

In delving into representations of gas pipelines, we essentially introduce the visual elements that translate them. The cartographic format will be explored as a document that encapsulates within its graphical and conceptual features the political, economic, or social objectives driven by various entities: a country, a company or a community. By revealing the specific interests behind the creation of a particular map and the context in which it is displayed, we can unveil the strategic and inherently subjective nature of cartographic representation. The gas distribution infrastructure in Europe has been depicted numerous times through network diagrams overlaid on geographical maps of the regions it spans. Their representation primarily comprises the routes of the pipes delineated by lines, accompanied by the depiction of connecting nodes in the background. These nodes, which usually signify extraction wells or gas delivery points like industrial hubs and cities, are often represented by dots. These are the primary graphical components, condensing the journey of gas into a tangle of lines.

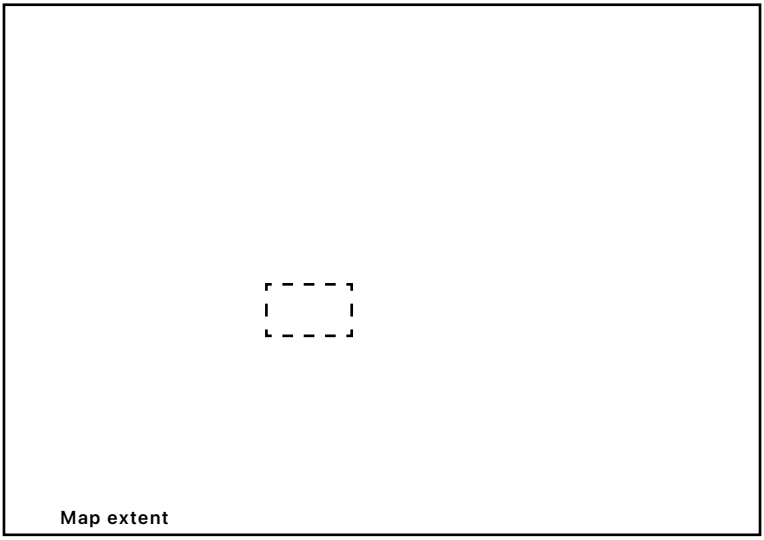
**IMG05**
Italian network of gas pipelines (1955).



Pipelines



Node points

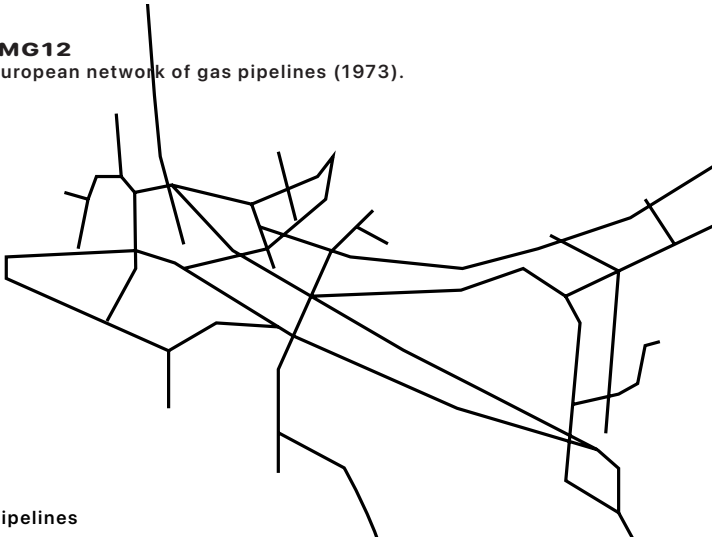


Map extent



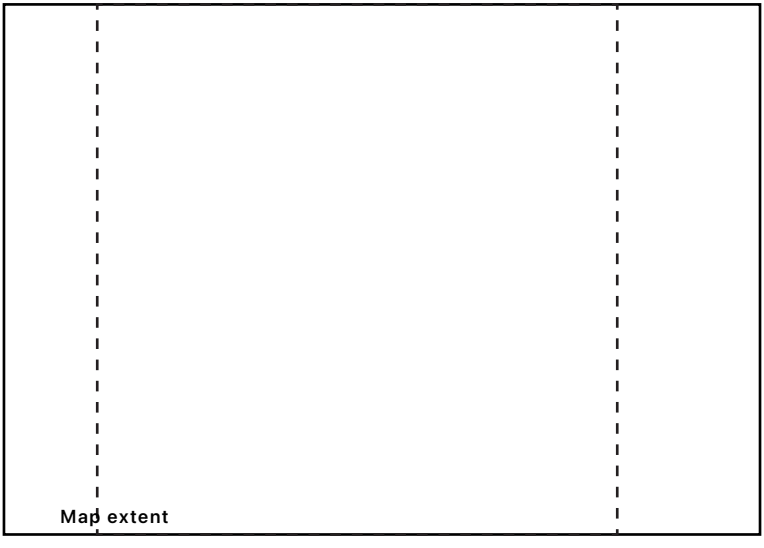
National borders

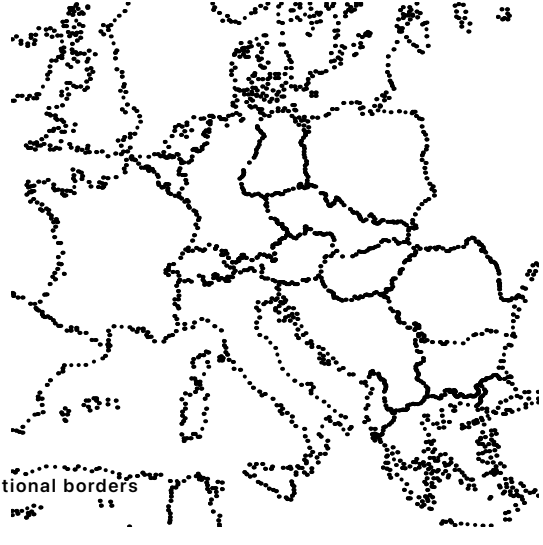
**IMG12**
**European network of gas pipelines (1973).**



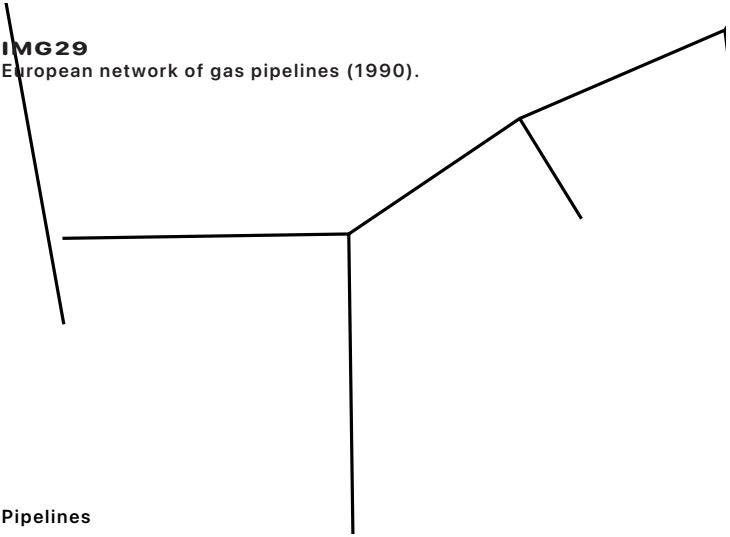
**Pipelines**

**Node points**


Map extent


National borders

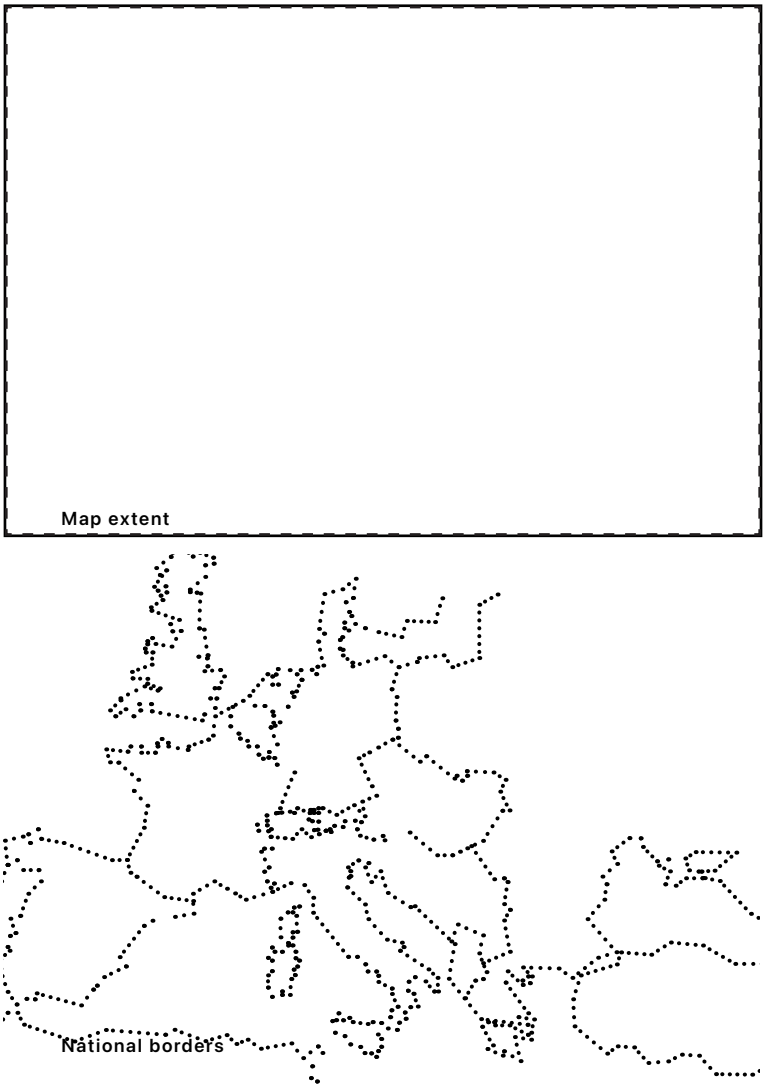
**IMG29**
**European network of gas pipelines (1990).**



**Pipelines**



**Node points**

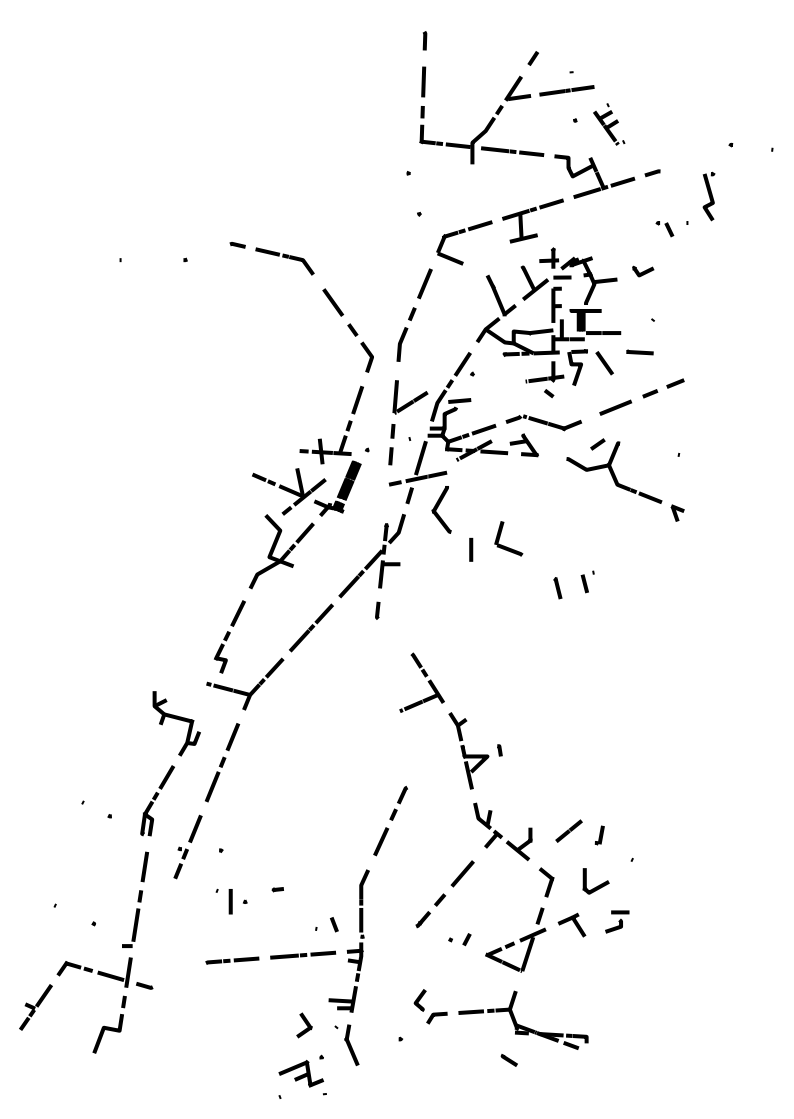

Map extent
National borders

Just like a schematization of a metro or rail network, the pipeline route is usually simplified to polygonal lines, smoothening their complex way to overcome or circumnavigate natural or man-made obstacles such as rivers, towns, and mountains. Other basic data-visualisation techniques are used to enrich the informations about the lines, such as stylizing it depending on state of construction or its thickness by diameter size and flow rate. Other features are key to visually translate extensive networks, such as the scale of representation, the map extent, the reproduction size and the medium used. **The act of representing the infrastructure through the cartographic format fixed it as a tangible and permanent object even if, as in the case of gas distribution, the object itself is relatively invisible to the general public and somewhat uncertain even to workers and decision makers**[7].

The design of publicly accessible representations and diagrams of this infrastructure begins simultaneously with its official commissioning. In Italy, for the first time in Europe, this moment occurs around the 1950s. The visual material used to tell this story will consist of the cartographic representations of the gas infrastructure network that it was possible to find, produced exclusively by the company in charge itself, again underlining the strategic aspect of cartography. There are approximately three main sources of material found and used, all dating back to the Italian state-owned company ENI: the archives of the Milan Trade

[7] Dodge, M., C. Perkins. (2010).

Fair (Fiera Campionaria di Milano), where we find a unique case where the gas pipeline network and its geography were presented to the public in the form of pavilions' decorative banners, as a product and symbol **IMG05**; ENI's corporate paper magazine "Ecos" **IMG13** where the work and expansion of the network was promoted in the form of informative diagrams and maps; and finally other materials comes from the most recent technical representations released by the companies themselves **IMG35**. The materials we will be dealing with are made of diagrammatic and approximative artifacts, very often far from cartography, abstracting the network of pipes to an image and symbol of technological achievement and expansion.



# SELF-SUFFICIENCY DREAMS CARTOGRAPHY AS PROPAGANDA

Natural gas is a resource that has been discovered relatively recently. It was not until the 1950s that extraction wells were officially opened in Europe, the first of which was in Italy. The north Italian sandy flat lands were a perfect terrain in which to find hydrocarbons, and in the frenetic rush to find oil, which was and kept being for decades the main national energy source, methane was extracted almost as if it was an inevitable implication of the latter. Methane back then was used very little and only very locally. The Italian state-owned company, today known as a multinational oil & gas holding, Eni, will build the then-largest European network of pipelines to transport this resource throughout the north and later to the whole country. Between the 1940s and 1950s, multiple wells were opened: *Cortemaggiore*, *Caviaga*, *Ripalta*, are the names of the small groups of farmhouses in the countryside of northern Italy that became the key spots to obtain this new "clean" resource. Indeed, gas at first was celebrated as an innovative and ecologic resource capable of replacing the more polluting oil and of which Italy and Europe speculated to have more availability. Based on these assumptions Eni will build an entire narrative and communication strategy throughout the years. I mention "speculation" because it was probably clear from the start that the quantities of gas at least within the Italian country could not be enough in the long term, not even to sustain Italian industries. Yet an endless network of pipes was silently laid under the feet of dreamy Italian citizens at the time of the economic boom.

In some of the maps describing the distribution of Italian gas in those early years, at the center of the network were precisely those towns like Cortemaggiore **IMG07** ■ , places hardly represented on any other map because of their very small size and their apparent strategic usefullness. In this kind of cartographies, however, they took space and remained marked as crucial nodes of the network until the most current gas distribution maps. Cortemaggiore, located in the north of Italy, is one key place where the history of gas and oil overlap and where the former replaces perfectly the latter. The small town became famous for the first time throughout the country because of the gasoline produced from its fields, publicized directly through its name: "Supercortemaggiore". The gasoline remained for sale for more than a decade, leading the company to build countless gas stations and related infrastructure together with it, pushing its promotional narratives. With the end of oil to be extracted, both the gasoline and the fame of this place disappeared. Cortemaggiore later remained a gas extraction site and still today is part of the invisible map of the distribution of this resource. The name of this little town, by being used to promote a product, exposed to the bigger public the geographical presence of oil in a specific location. Even if today this choice sounds weird (because we would never know the provenance of the gas or oil we are using) this specific story comes from a nationalistic communication strategy common in those times to emphasize the domestic provenance and thus its inherent quality.

One of the first maps representing the gas network and captured in a photograph is from 1950 **IMG01** ■.

A wall a few meters high of an exhibition hall displays a map in a frame. The representations of the methane gas distribution network until at least the late 1950s are all found in the Milan trade fair archive and always in the bizarre format of 'architectural' building-size maps **IMG08** ■.

The fair was an exhibition open to all, with the aim of presenting the innovations of the most important Italian and foreign industries. However, it was not only a stage for industrial innovation, but also and above all a showcase for contemporary advertising, architecture and (graphic) design. Cartographic-like representations took up a large space on the facades of Eni's pavilions for a long period, acquiring a propagandistic role and losing the purely informative one. We find these network diagrams on the buildings of Eni's group until the sixties. Their geographical extension is still very small and features just the northern part of the country where gas and oil have long been known to be located. The relatively small extent of the infrastructure is amplified by the size of the walls on which they are applied. **The insistence on using the methane pipeline network as image and symbol, both as covering and as advertising content, was probably aimed to be a part of a larger strategy, which might seem absurd today, to make visible what was invisible to everyone, namely the gigantic network of underground pipelines**[8]. We note, however, how this way of 'showing' actually involves reducing the infrastructure to a symbol, a drawing,

[8] Deschermeier D. (2009). Impero Eni. L'architettura aziendale e l'urbanistica di Enrico Mattei, 130.

an approximation. The external view of the Eni pavilion in 1954 **IMG01** ■ shows a more enlarged and extensive diagram of the gas routes, supported by the slogan: "from the depths of the earth, energy for Italian labor", a phrase with a nationalist and propagandist slant that highlights a contradiction. Even if the network depicts only the northern part of Italy, the absence of any other topographical and political layers, such as national and regional borders, morphological elements such as mountain ranges or coastlines, makes the viewer more indifferent to the map's national incompleteness. The interplay of the network diagram, the slogan, the company logos and a three-dimensional graphic element on top of it makes the façade of the building look more like an advertising poster than a cartography. But this representation does not appear so strange if we contextualize the era in which it was created. In the 1950s, Italy was a country in the midst of an economic boom, where industries and their technologies were a symbol of national power and pride. A capillary network of pipes underground was at the time a revolutionary and innovative idea, capable of providing a new, powerful and 'clean' resource over a vast area, overcoming obstacles and distances. It was around the cities of the north that the industrial centers were increasingly in need of energy, also given the growth in exports and the need to rebuild infrastructure destroyed during the war.

**The prevailing industrial and agricultural productivity in the north of the country are clearly visible from these early maps**[9], where the network only stretches its tentacles as far as cities in the north, without ever continuing further. **But also the absence of the rest of**

**the country in these representations comes from the extent of permits given to the company to explore underground, which at the time was just the Po Valley area**[10]. It was therefore the only company that could create such an infrastructure without other competing enterprises and with almost unlimited availability. Industry, especially back then, was the main destination for natural gas. The slogan printed together with these maps is proving that "labor" is the main target of this infrastructure. Heavy industries such as steel mills benefited from large quantities of heat produced by methane. Famous were the steelworks and factories in the outskirts of Milan and Bergamo, the first to be reached, both because of their importance and proximity to the extraction sites. The trajectory of pipes was running around big urbanized areas and therefore reaching its industrial hubs rather than residential neighborhoods. Gas as a domestic energy source, delivered from house to house, will arrive later, replacing the narrative of industrial power with that of comfort and modernity. The frivolity of these public announcements perhaps also masks the limited use of this resource at the time: gas, which had just been discovered, had no clear future or perspective. Its production and consumption could not compete with the massive use of oil to supply the country. **Gas represented a new dream to believe in, the dream of a clean, powerful and homegrown energy resource.**

[9] Catino F. (2013). Il Contributo italiano alla storia del Pensiero.
[10] Ibid.



# THE LONG ROUTES OF ENDLESS SUPPLY CARTOGRAPHY AS A NARRATIVE TOOL

It's the 1970's; the maps of the gas network are no longer used to cover pavilion's facades but instead get smaller in scale and representation, filling the pages of the newly company magazine. Their more designed and playful graphic language is used to support articles that narrate the company's new values: diversification, internationalism and technological development. And to do so, the cartographies adapt, showing where the new world ends.

On the one hand they depict the construction of the first import pipelines from the Netherlands and former USSR, and with them an entire set of images and representations. On the other they speculate on the consequent creation of a European grid, presented to Italians as a huge effort to provide them the dream of "endless" energy supply.

On 8 June this year were [...] officially inaugurated the two international methane pipelines USSR-Italy and Netherlands-Italy. Two important achievements with which Eni guarantees, in the difficult energy situation we are going through, an adequate satisfaction of the growing demand for methane that our country expresses and which by now exceeds even the significant national production. Even a few statistics give the exact measure of the organizational and technical effort required. 774 km of large-diameter methane pipeline to transport the gas produced in the USSR from the Czechoslovak border to Sergnano, in the heart of the Po Valley. A further 826 km to bring Dutch gas to Mortara, close to the large industrialized areas in the north. [...] The total investment amounts to about 480 billion lire, 25 percent of which is for works carried out in Italian territories. [...] Beyond the significance of a far-sighted energy policy initiative (the agreement for the construction of the methane pipelines in fact predates the outbreak of the oil crisis), this achievement constitutes an example of cooperation that consolidates and enhances the relations that Italy entertains within the European Economic Community, on the one hand, and with the Soviet Union, on the other.

Extract from an article on Ecos 1974

These are extracted pieces of an article published in 1974 in Eni's brand new magazine called Ecos, founded two years before to inform employees and a more general public of projects and stories related to the company. From the seventies up to the nineties all the cartographic material found regarding the gas infrastructure is communicated through this editorial format. Published in Italian and English, it was distributed free of charge in Italy and abroad to the employees, people from politics, economics, culture, journalism, and municipal and school libraries.

Although it had the appearance of a magazine full of technical details, Ecos actually looks more like telling stories around company projects, often not even directly related to them, with a strong narrative tone. Ironically, the name of the magazine itself was chosen to remind both the words "Economy" and "Ecology" unveiling the main narratives deployed and pursued. Even though the building's decorative "maps" belong to other times, also in the magazine we find many cartographic drawings of the network, but with new scales and details.

A map from 1973 **IMG12** ■ shows the European pipeline network in its earliest stage. The frame in which it is enclosed is a particular one, 'central/eastern' European, cutting out countries that we are used to see represented together with the rest of Europe. During these years, the Eni group planned and then built the "Olanda-Italia" connection line and the one with Austria to access the already existing line from the USSR. In addition, the gas pipeline from Algeria to Sicily was planned. The borders between countries drawn in this map are a clear indication of the political moment in Europe, showing a very different lay-

out from the one we know today. Germany is still divided between east and west; Yugoslavia and Czechoslovakia still exist as single entities; beyond Poland there is only one bloc without any division. Yet the thin national border lines are overlaid by the more pronounced black lines that trace the paths of the pipelines. More visible than these in hierarchical order are only the natural boundaries between land and sea. The gas pipelines therefore draw new borders and relationships, shaping an alternative geography: France, Belgium, Holland and West Germany are connected; Italy, as well as Austria and the United Kingdom appear as separated systems. It is the new lines under construction (in red) and planned (in blue) that will cross those still respected borders, creating a new geographic format that includes in the European context some North African countries such as Algeria, Tunisia and Libya, still today key places for the energy supply of Europe. The lines in the direction of the former USSR stop at the border with Czechoslovakia, giving no further details on the status of the line, probably already built as far as Vienna by then. It is the beginning of a geographical shift that starts to decentralize from Europe and moves outside its borders, even more distant from sight and perception.

In 1974 the construction of one of the above-mentioned pipelines to import gas to Italy, the one from the Netherlands, was completed. But why this happened? Why the domestic, italian and very high quality gas was not good anymore? As a consequence of the international energy crisis of those years, and the price rise of oil, gas began to take hold in many European countries as a primary energy source. Until then gas in Italy especially, although promot-

ed as a key resource, was by far secondary compared to other sources, **a sort of future solution not yet fully developed. All these maps never put gas into perspective with other energy systems**, but just fix it as an isolated system, not connected to others and able by itself to fulfill national or international demand.

In the Netherlands the discovery of natural gas deposits in the subsoil of Groningen in the late 1950s generated, similarly to Italy, a rapid process of building a capillary network to sell it. The quantities of gas were large and so from the very beginning Dutch gas was exported to neighboring countries such as Germany, Belgium and France, positioning this resource as a medium to long-term solution. In Italy, on the other hand, before the 1970's the network remained exclusively national. In the map found on the pages of ECOS **IMG12** ■, an international network of interconnected northern European countries is already visible. Dividing these two scribbles of lines is not only a border but also a significant natural obstacle such as the Alps. Work on the construction of the pipeline to northern Europe was often proudly portrayed in the company magazine in those years and the following decades, through reportage photographs. Underground tunnels **IMG19** ■, snow-covered forests, lofty peaks, sun-drenched countryside, river crossings; this is the imagery used to describe a work that was historic at the time, that of connecting northern Europe with Italy with a line almost 1000 km long, crossing different environments and challenges: from the heights of the Alps to the negotiation of the passage of private property along the way. In these photos, the pipeline is depicted in relation to the landscape in which it will fit.

Gutted forests or plains furrowed by rectilinear potholes disappearing into the horizon are images often used in this magazine to show the power and technological advancement of a project that seemed impossible to realize: today, those same photos seem to come from distant times.

"Imports of natural gas from the Soviet Union and the Netherlands realize in practical terms those energy policy goals that our country has been pursuing for some time and to which the country and the government are now fully committed and must commit themselves with ever greater awareness and incisiveness. With these imports, Italy's overall energy supply is increased; a geographical diversification of energy supplies is achieved: a diversification by energy source is implemented, containing the sovereign weight of oil in Italy's energy balance: the availability of a 'clean' and particularly useful source of energy for certain uses is increased. These objectives are then fulfilled at a notoriously delicate time in the energy situation. They are especially important for us, for Italy, which must address and solve the problem of its energy needs in more complex and peremptory terms than others, if we want to sustain and develop its production apparatus and remain among the great industrial powers. [...] The massive project we are inaugurating today gives us a significant example of the conquests of which technology is capable, if it is correctly and usefully placed at the service of man and peace: it constitutes the most formidable way of bringing peoples together. The great railways built at the end of the last century played a decisive role in making Europe a community of nations, despite the delays and setbacks that, unfortunately, occurred at the political level. So today, if we think of the irreplaceable and decisive role that energy has and will have in the development of relations between states, it is clear to look at the two pipelines as the concrete symbol of ever-closer international cooperation on a directly operational level."

Extract from an article on ECOS 1974, speech from the former Italian prime minister Mariano Rumor.

The objective behind the construction of the pipelines from the Netherlands and Russia, to use the words of politicians from then and now, was 'diversification'. Diversification was happening because of the international energy crisis, which pushed gas from its potential usage to its full scale deployment. But was also a geographical diversifica-

tion, replacing Italian domestic production with production from afar. Italy therefore, even though it still produced gas domestically, with the construction of these pipelines irreversibly connected itself to a more extensive European network, the one that is today dependent on resources outside the European Union.

The Netherlands gas pipeline, unlike the Russian one, is not only presented as an import line, vanishing at the edge of the pages to unknown and distant places **IMG24** ■. It is often narrated as a distribution backbone between the countries it passes through, depicted with its entire route between Italy, Switzerland, Germany and the Netherlands. The gas pipeline from the Netherlands fills pages and pages of the company's magazine, with stories more or less related to the pipeline itself, such as compelling tales from the construction site or travel stories along the landscapes through which it passes. The one coming from the USSR border doesn't take as much space and even when it's the main subject of an article, the focus stops on the big works employed to cross the alps, never going geographically further.

The two pipes however, also because built almost at the same time, were usually associated to the same strategy, losing the specificity of each of them. The need to show these new international lines led the cartographies of these years to take two formats: one, that shows the entire system, loses details and removes information by no longer presenting location names, cities, extraction points **IMG29**■. Another type of cartographic representation, on the other hand, still prioritizes the national dimension, this time encompassing its entire territory but effectively re-

moving its connection to the rest of the system IMG26 ■.

Places of import, which will increasingly become key, are indicated by simple dots on the map or arrows defining the direction of the transport. The scale of the infrastructure, by expanding across national and continental borders, needs to be scaled up or down, by reducing details or by framing single parts of it. In either case, what it's enabled is a progressive invisibilization of how, why and where the network is operating.

# REVERSE FLOW CARTOGRAPHY AS A SILENT DOCUMENT

Today the abstraction of the gas distribution infrastructure is at its peak. What is visible and detectable of it today is very different from what we have seen in the almost 70 years of using this resource in Italy and other parts of Europe. At the Milan trade fair, which still exists albeit under a different name, we are unlikely to find Eni proudly presenting the network of underground pipes transporting gas through maps and diagrams. On the other hand Eni still publishes a company magazine, but under a new identity and in the name of a new dream. Today it is called 'World Energy' and inside we no longer encounter maps of the gas infrastructure, but rather informative articles on other technologies and resources that make us imagine better futures; Hydrogen, solar energy, wind power plants and other imaginaries that exists just either at a planetary ("world") scale or in the future. **Not only it have become opaque the location and structure of its flow, but today even the very direction of gas movement.** The European gas distribution system has now become an 'interchange' network, in which some routes have reversed their original direction. Gas is now also put into circulation by countries which were mainly importing before, like Italy, after buying, storing and then re-selling gas on demand, thus losing track of the original geography of the resource. It is often called '**reverse flow**' in the newspapers and means that pipelines that were planned in one direction are now usable in both.

The place where we find visual material about the gas distribution network is today probably only the web, where many maps sketched by newspapers and news outlets focus on single topics, such as a country's import and export or a pipeline under construction or planned. Maps are no longer just made and designed by the company operating the infrastructure, yet those created by third parties seem to lack detail and information.

Looking specifically at Italy, where this story began, the most recent map we find is from 2015, downloaded from the SNAM website (a sub company of Eni) after a long search. Displayed by regional macro frames, it depicts the route of the pipelines with a fairly high degree of detail, allowing us to understand roughly where the pipeline passes. Another map **IMG35** ■, also released by SNAM in 2013, instead represents the national network and its entry and exit points to foreign countries. This one too, with a similar graphic style, where none political or physical cartographic elements appear, is limiting to understand the distribution network. This map, however, representing an infrastructure that hardly changes over time but only expands, invites us to look for the names of those places in the flatlands of Northern Italy where gas was surprisingly found and extracted in the late 1940s. Cortemaggiore and Ripalta are present on the map and, in different ways, still represent key nodes for the current network. The natural gas in the subsoil of these places has been completely exhausted and, even if there was any left, it would probably cost too much to extract it. These, as well as other fields exploited in the early years of the Italian "metanizzazione", are now storage points used to guarantee a constant gas flow in the national

grid without the risk of running out. On the one hand, this process stabilizes supplies by gaining a certain degree of independence on exporting countries; on the other, it cancels out the impact of any climatic (a long winter) or geological (a shortage of extracted gas) event by abstracting the resource to a product without provenance or history. Once again, the illusion of an endless flow of ready-to-use energy is pursued.

The national pipeline map in pdf format, the appearance of which seems so official and bureaucratic in its total absence of any stylisation (colors, headings and typographical elements) makes us think that it was not created for any communication strategy but rather is the result of a legislative obligation, that of making public some limited informations about the infrastructure, so that it does not disappear altogether. **The critical analysis of these network maps can provide important insights into the objectives of a state or company in conveying the perception of infrastructure**[11]. Perhaps behind the unstylised nature of this cartographic piece lies another strategy to turn the gas distribution network into a mere technical-scientific object to be ignored.

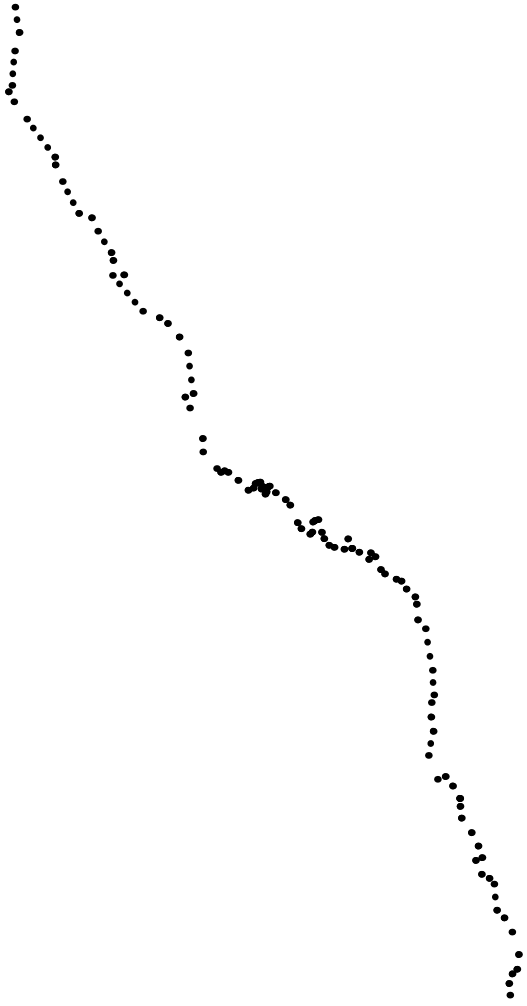
[11] Krygier, J. B. 1997. Envisioning the American West: Maps, the Representational Barrage of 19th Century Expedition Reports. Cartography and Geographic Information Systems 24: 27–50.

**The Netherlands-Italy gas pipeline, built to connect a country that was short of gas supplies to one that seemed to have many, is today one of Italy's least utilized methane import lines**[12]. In 2022, the direction of the flow was also reversed, outlining a completely new scenario: the Italian export of gas across the Alps. In 2011, the Eni group sold all its shares of this gas pipeline. Thus, the long path proudly built and promoted by the Italian company in the 1970s and 1980s was left to other holdings. Although the pipeline has been promoted as a single line, it has always been fragmented into two main sectors, on the two main countries it passes from: Transitgas (in Switzerland) and Trans European Naturalgas Pipeline (TENP, in Germany). Today, when talking about pipelines, media coverage focuses on those that currently generate tangible problems for Europe and those that are important in sustaining Europe's energy needs, such as lines from Russia and North Africa. On the other hand, indifference towards other lines such as the one between the Netherlands and Italy has increased due to its current lack of strategic importance. Yet this pipe somehow still retains a role in the complex system of internal European exchange, but by changing its meaning, its direction and even giving room for speculation about possible future uses. Indeed, one can already find reports issued by various gas companies which foresee the conversion of some pipelines to transport hydrogen **IMG36** ■, the solution

[12] Treccarichi, C. (2023, February 13). Quanto gas abbiamo per l'inverno. Today.

that is today believed to be the most innovative and the cleanest. It's the same dream of sixty years ago but materialized in a new element, a new shape and a new image.

**This pipeline, in its inherent invisibility on the territory, surrounded by media silence due to its lack of relevance and camouflaged by the landscape over 50 years after it was laid under the ground, is now totally invisible and therefore a symbol and 'relic' of a past where this line was covered up by narratives and transformed in the source of desires, which are now old and outdated.** Its history and the reasons for its construction, parallel to those of the gas pipeline from the former USSR, speak of energy needs that the European territory could never sustain; they speak of a diversification that just partially took place; they speak of an energy empire built outside national borders. As with all stories, this one too comes to an end. In the dark night of European energy supply, it is time to go back to sleep and chase a new dream.



# CONCLUSION

After scrutinizing these graphical artifacts meticulously, one after another, for months, a gradual process of concealment and erasure becomes apparent. The maps, with their lines and arrows, represent a growing indifference toward a resource and its entire transportation infrastructure: from initial public promotion with nationalist and autarkic narratives, to the crafting of informational artifacts to cultivate new imaginaries and dreams surrounding the expansion of lines beyond Italian borders, all the way to the current neutral and bureaucratic design of documents. This process of camouflage and waning interest in a resource and its infrastructure seems to unfold simultaneously with its real-world, large-scale implementation. The dream remained untouched until gas emerged as a pivotal asset and a major national and European energy source.

Similarly today, the European pipeline network is hardly represented when talking about gas, because of its importance and political meaning. Whereas the same network, or at least part of it, is drawn in colors on reports presenting its future conversion to hydrogen. This case study shows a specific historical and geographical context that tells us how technology was promoted and presented to the public during the 20th century. Perhaps today even the most innovative and promising of infrastructures would never be presented through its geographical locations and paths as a huge banner on a building or a spread

in a magazine. Maybe invisibility has become part of the language of infrastructure. Because of the ever-increasing change of desires and jumping from one dream to another, we have learnt not to show and abstract these networks more and more.

However gas is one of the multiple energy resources that had a similar life cycle. Today the European energy supply, even if highly based on fossil fuels, consists also of different systems to cover its high demand: the more established and not promoted hydro-power; the enigmatic and controversial nuclear power; and the newer and desired renewable energies such as wind or solar power, which together with the above mentioned hydrogen render the next future scenario. Each of these has a specific infrastructure that supports and distributes its energy, and together with it a representational sphere made of images, diagrams, maps, that sparks a specific imagery and idea, like the one we explored with gas. What this archival material shows is on one hand the separation of these different systems when it comes to their representation. They are usually treated as unique and isolated infrastructure able to provide enough energy on their own. What happens is that these are usually intertwined, not only conceptually but also on territory, especially when an infrastructure like the gas one takes so much space and connects two opposite corners of a continent. On the other hand it shows how the related imagery is proportional in quality and quantity to the hopes and desires it carries.

Ultimately, the diverse narratives surrounding this infrastructure not only address gas but also outline the evolution of the European Union and its political ties with

member states and neighboring countries. Cartography, and graphic design more broadly, have provided me with enough elements, albeit chaotic and fragmented, to compose a specific yet plausible portrayal of how our world has been shaped and could evolve. What we possess are fragments, extracted almost incidentally from maps and diagrams never intended to depict history. The persistent need to construct new imagery around emerging technologies or resources, integrating them into a constantly shifting collective dream, exposes the fragility and transience of our understanding of the world. Cartography in itself, like other visual representations, proves to be a mirror of historical scenarios and a document situated in time and culture. And at the same time it is a tool that has little to do with the neutrality and objectivity with which we associate it.

Although this text ends here, this research has led me to another chapter of this story: a fourteen-days-long journey along one of the pipelines mentioned in the previous pages, the one connecting Italy and the Netherlands. A travel to look for what is left of this abandoned dream, or if it has even ever existed. The first approach to this topic was as well a very close look to this tube, without fully recognizing its language and signs. After these pages and these months, the same territories and spaces I thought to look at, have assumed entirely new significances: Those yellow little sticks in the ground are the gas trail marks, there to grab your attention and at the same time designed to keep unwanted observer far away; a linear gap amid a forest is a secret passage, a protection and an open access for maintenance of a pipe; a cryptic concrete building on

an hiking path is a measuring station but also the door of a long mountain tunnel; a little fenced citadel where I'm kindly asked to go away when I take pictures, functions as a pumping station, where gas is measured and distributed accordingly to demand. These places exist probably with similar volumes and shapes in different countries, around where we live, where we go on holiday or where we work. These objects are meticulously crafted to evade observation, purposefully engineered for invisibility. All these spaces talk about the life of the infrastructure and the work behind its functioning. Spaces that are not fully visible from a map and neither from the ground, but just by moving continuously between the two.

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| **IMG01** | Agip Pavilion at Milano Trade Fair 1950. Source: Archivio Storico Fondazione Fiera Milano. Author: Publifoto. |
| **IMG02** | Enrico Mattei, former president of Eni, at the first national gas conference. Piacenza, 1952. Source: Archivio Storico Eni. Author: unknown. |
| **IMG03** | External view on the Eni group pavillion at Milano trade fair 1953. Source: Archivio Storico Fondazione Fiera Milano. Author: Publifoto. |
| **IMG04** | External view on the Eni pavilion at Milano trade fair 1954. Source: Archivio Storico Fondazione Fiera Milano. Author: Raimondo Niccolini. |
| **IMG05** | External view on the Eni pavillion at Milano trade fair 1955. Source: Archivio Storico Fondazione Fiera Milano. Author: Ancillotti and Martinotti. |
| **IMG06** | External view on the Eni pavillion at Milano trade fair 1957. Source: Fondazione Fiera Milano. Author: unknown. |
| **IMG07** | Italian gas pipeline network on GattoSelvatico 1959 06b. Source: Archivio Storico Eni. Author: unknown. |
| **IMG08** | External view of Eni pavilion at MIlano trade fair, 1961. Source: Fondazione Fiera Milano. Author: unknown. |
| **IMG09** | Frame from "L'Italia non è un paese povero", 1960. Source: Archivio storico Istituto Luce. Author: Joris Ivens. |
| **IMG10** | External view on the Eni pavillion at Milano trade fair 1962. Source: Fondazione Fiera Milano. Author: Publifoto. |
| **IMG11** | External view on the SNAM pavillion at Milano trade fair 1968. Source: Fondazione Fiera Milano. Author: unknown. |
| **IMG12** | ECOS 1973 11-12. European network of gas pipelines. Source: Archivio storico Eni. Author: unknown. |

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| **IMG13** | Cover of ECOS magazine 11-12, 1973. Source: Archivio storico Eni. |
| **IMG14** | The Netherlands-Italy pipeline construction site between Germany and Switzerland, 1973-74. Author: Agenzia Giornalistica Italia. Source: Archivio storico Eni. |
| **IMG15** | ECOS 1974 015-016. Launch of the gas pipelines from the USSR and the Netherlands in S. Donato Milanese. Source: Archivio storico Eni. Author: unknown. |
| **IMG16** | The Netherlands-Italy pipeline construction site between Germany and Switzerland, 1973-74. Author: Agenzia Giornalistica Italia. Source: Archivio storico Eni. |
| **IMG17** | The Netherlands-Italy pipeline construction site around Mottarone (Verbania). 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG18** | The Netherlands-Italy pipeline construction site around Mottarone (Verbania). 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG19** | The Netherlands-Italy pipeline tunnel crossing the Alps, 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG20** | The Netherlands-Italy pipeline construction site on the Alps, 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG21** | The Netherlands-Italy pipeline construction site and visible path in Brienz (Switzerland), 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG22** | The Netherlands-Italy pipeline construction site between Switzerland and Germany, 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |
| **IMG23** | The Netherlands-Italy pipeline construction site around Mottarone (Verbania), 1973. Source: Archivio storico Eni. Author: Agenzia Giornalistica Italia. |

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| **IMG24** | ECOS 1976 040-042. Gas import lines to Western Europe in 1975. Source: Archivio storico Eni. Author: unknown. |
| **IMG25** | Interiors of the Eni pavillion at Milano trade fair 1978. Source: Fondazione Fiera Milano. Author: CPF Novitype. |
| **IMG26** | ECOS 1989 189-191. Italian pipeline network at 1st January 1989. Source: Archivio storico Eni. Author: unknown. |
| **IMG27** | ECOS 1991 218-220 C. Italian pipeline network at December 1990. Source: Archivio storico Eni. Author: Studio Ruffolo. |
| **IMG28** | ECOS 1991 218-220 C. Natural gas deposit in Italy, Europe and Northern Africa. Source: Archivio storico Eni. Author: Studio Ruffolo. |
| **IMG29** | ECOS 1994 060. European gas pipeline network. Source: Archivio storico Eni. Author: Unknown. |
| **IMG30** | ECOS 1991 218-220 C. Snam dispatching centre control room. Source: Archivio storico Eni. Author: Unknown. |
| **IMG31** | Gas pipeline ground marker nearby Groningen (Netherlands), 2022. Source: Google Street View. |
| **IMG32** | Ground marker nearby Eindhoven (Netherlands). Source: personal image. |
| **IMG33** | Ground marker visible from drones and helicopters nearby Brienz (Switzerland), 2023. Source: personal image. |
| **IMG34** | TransitGas pipeline trajectory, 2021. Source: Fluxys. Author: unknown. |
| **IMG35** | Italian national gas pipeline network, 2013. Source: SNAM. Author: unknown |
| **IMG36** | Map of future European hydrogen pipelines using old gas ones, 2022. Source: European Hydrogen Backbone. |

